Farkle Implementation Part Three, Fiona Callahan

Goal Of My Program

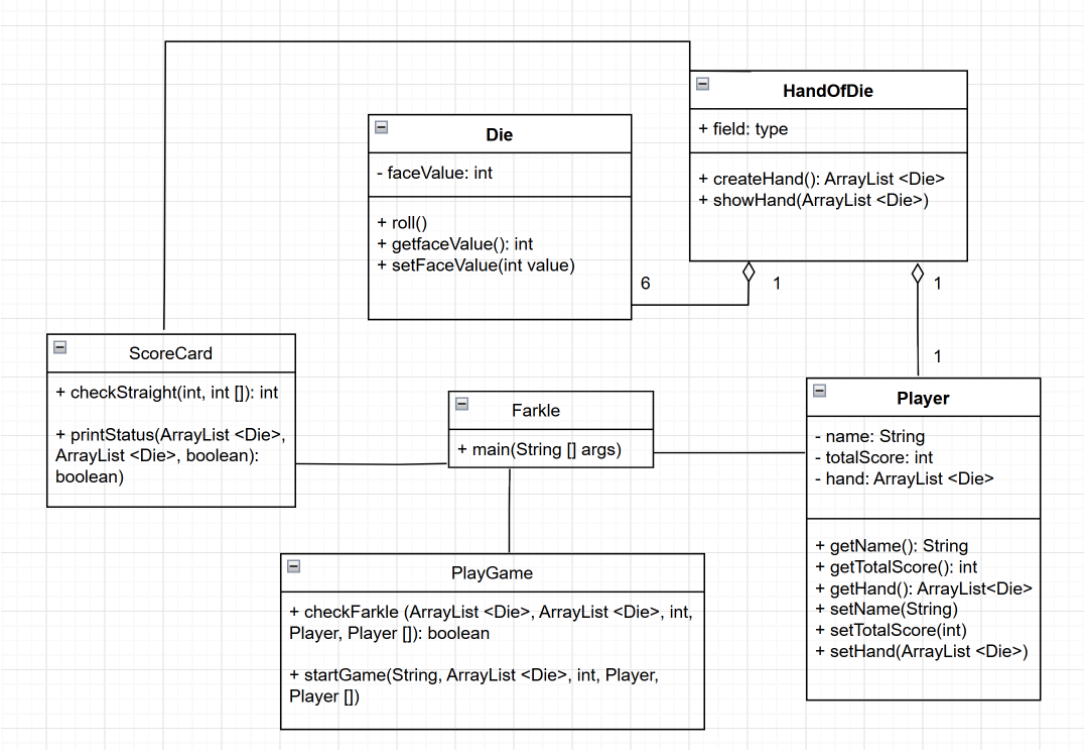
- My program implementation, “Farkle”, models the real game that includes elements such as dice, the hand, scorecards, the current player, and melds. It allows a multi-player game interface. The players can roll their dice, and decide to move certain dice into the meld. For the dice that they keep in their hand, they are able to reroll. One new feature added in this program is the ability to detect a HotHand, however, the user must beware because a Farkle is still possible. If the user decides to quit, they do not gain their melded points, and instead, just are returned their total score. Another few features added in involve a FullHouse detection, score updates per turn, and a proper Farkle-style endgame implementation. The main goal of the program is to make it user friendly, and make sure there are clear steps forward at every point.

General Design of the Program

- The design I chose for this program includes five different classes: Scorecard, PlayGame, HandOfDie, Player and Die. I chose this design because it blocked each set of functions off by way of similar instructions. I followed Java’s OOP style of programming. The Farkle class is what kicks off the program by creating a hand before it sends the first user to “PlayGame”. PlayGame is what I see as the main, as it includes the method, checkFarkle, which leads the user through the roll. The Die, and the HandOfDie class go hand in hand because an object, hand, is full of the die objects. I oriented my program around ArrayLists, as I found this easiest to access variables and keep changing its length and values. The user is always shown different options for steps forward, and prompted when appropriate. The meld score is displayed at the bottom of each menu, and the total score is displayed at the end when the user is done with the game. Depending on whose turn it is, the program keeps track of the player’s name, total score and hand. There is also an option to “play to” a certain score. This means that it can go on for several rounds before it quits based off this condition, or if a player quits on their own.

Unit Test [Disregarded this section per Wang’s request]
****will go over this in next few classes**

UML Class Diagram



UML Sequence Diagram [ignored per Wang's request]
****will go over this in next few classes**

Major Design/Programming issues

- The main design program that I found while coding Homework 3 was deciding how I wanted to switch players when it reached the end of a turn. This led to the development of a Player class to hold their total score, name, and their hand. The hand variable is not needed due to the fact that it gets dynamically created every turn, but it does hurt to have it implemented. I also had several programming issues regarding the EndGame logic. I needed to figure out several booleans to indicate when someone had another turn vs when the game should be terminated. To overcome these challenges, I had to think through the game flow, and perform a lot of trial and error situations.

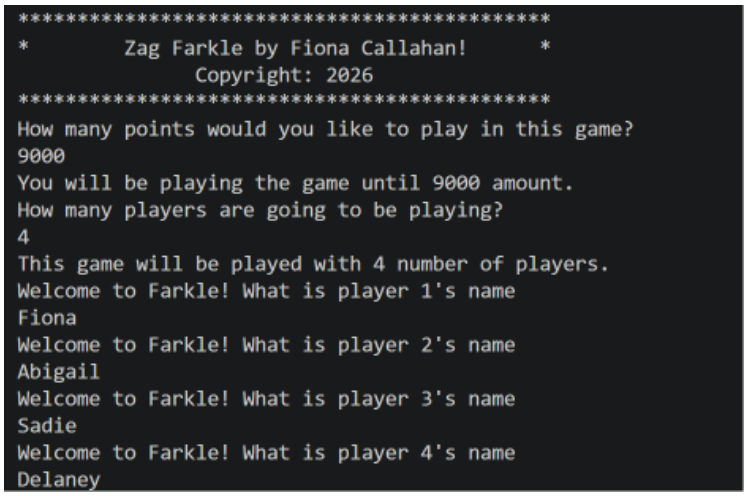
What would I have done differently

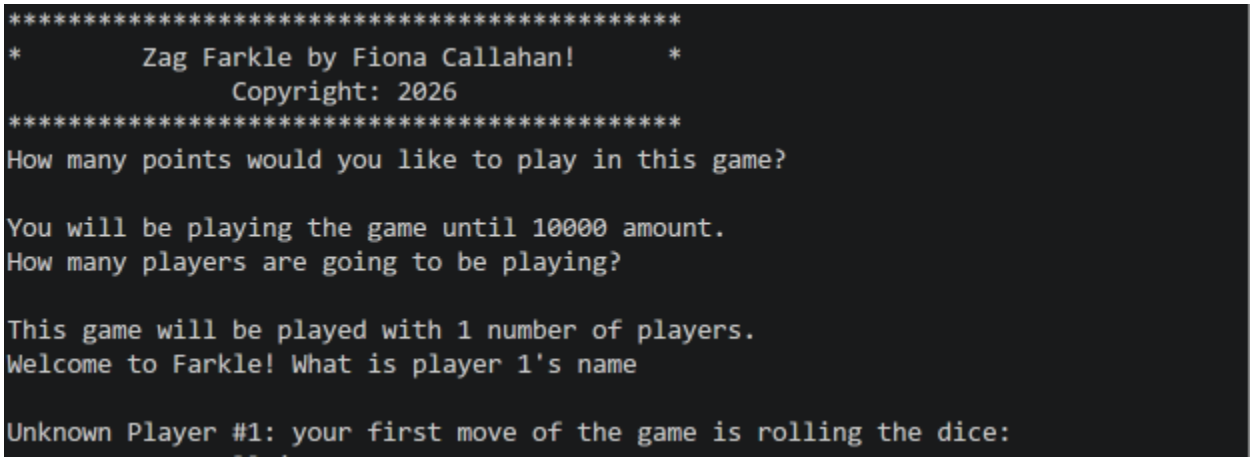
- I especially enjoyed coming back to this homework this time around because I was able to implement a lot of the features that I mentioned in this part from Homework 2. Due to the implementation of the Player class, it seems to run a lot smoother. However, there is still a lot that I would like to change for next time. First, there is an opportunity to include several more classes to allow the segmentation of the program more efficiently. I also wish I could rework the "Hot Hand" section. From homework two, this seems a little

clunky in its logic, and next time, I would create a separate class so that when a hand could be classified as a Hot Hand, it would be easier to move forward with. Finally, if I had more time, I would mess around more with the user interface. I want it to look cleaner. If I had more knowledge on user interface tools, I would try and make this more so like a real Farkle game. When I obtain these tools, I plan to come back and see how I can apply them.

Appendix [screenshots]

Intro screen with the Requirement #1 game menu





A shot of a Full House combo working

```
************************ Current hand and meld *****************
 Die   Hand |   Meld
-----------+--------------
 (A)      |    1
 (B)      |    1
 (C)      |    2
 (D)      |    2
 (E)      |    2
 (F)   2
  |
--------------------------------
               Meld Score: 1500

 (K) BanK Meld & End Round
 (Q) Quit game
(R) Reroll the dice not yet melded
                                 K
Enter letters for your choice(s):

ROUND OVER. total score is now: 1500
```

A shot of someone winning the single player game from Requirement #2

- (sneaky move: set the points total to 1000 and play a short one)

**screenshot down below was from when the game was played to a total score of 900

```
***************************** Current hand and meld ******************
 Die   Hand |   Meld
-----------+--------------
 (A)      |    1
 (B)      |    1
 (C)   6
  |
 (D)      |    1
 (E)      |    5
 (F)   4
  |
---------------------------------
               Meld Score: 1050

 (K) BanK Meld & End Round
 (Q) Quit game
(R) Reroll the dice not yet melded

Enter letters for your choice(s): K


Current Total Points:
    Fiona: 1050
ROUND OVER. total score is now: 1050
```

If you've done the Multiplayer game, shots of entering multiple players, the total scores, and the winner's screen.

```
*****************************************
*        Zag Farkle by Fiona Callahan!      *
              Copyright: 2026
*****************************************
How many points would you like to play in this game?
1000
You will be playing the game until 1000 amount.
How many players are going to be playing?
3
This game will be played with 3 number of players.
Welcome to Farkle! What is player 1's name
Fiona
Welcome to Farkle! What is player 2's name
Delaney
Welcome to Farkle! What is player 3's name
Sadie
Fiona: your first move of the game is rolling the dice:
```

```
Current Total Points:
    Fiona: 300
    Delaney: 1500
    Sadie: 500
ROUND OVER. total score is now: 300

Current phase of game play: ENDGAME
-------------------------------------------------------
Delaney has won the game with a total of 1500 points.
-------------------------------------------------------
```